

ONLINE BANKING INTERFACE

PROGRAM OUTPUT SCREENSHOT

The screenshot displays a Java Swing window titled "ONLINE BANKING". It features two text input fields: "Recipient Account No." with the value "12 3456789012" and "Transfer Amount (Rs):" with the value "5000". Below these are four buttons: "Transfer Funds", "Balance Inquiry", "Transaction History", and "Clear". A label shows the current "Balance: Rs50,000.00". At the bottom, a "Transaction History" list contains two entries: "Transferred Rs5,000.00 to A/C *****9012" and "Transferred Rs2,500.00 to A/C *****4455", both marked as "Successful".

SOURCE CODE

```
import javax.swing.*;
```

SOURCE CODE

```
import javax.swing.*;
import javax.swing.border.EmptyBorder;
import java.awt.*;
import java.util.ArrayList;
import java.util.List;

public class OnlineBankingInterface extends JFrame {
    private final JTextField accountField = new JTextField();
    private final JTextField amountField = new JTextField();
    private final JLabel balanceLabel = new JLabel("Balance: Rs50,000.00");
    private final JTextArea historyArea = new JTextArea();
    private double balance = 50000.00;
    private final List<String> transactions = new ArrayList<>();

    public OnlineBankingInterface() {
        setTitle("Online Banking Interface");
        setSize(650, 450);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);
        JPanel main = new JPanel(new BorderLayout(12, 12));
        main.setBorder(new EmptyBorder(18, 18, 18, 18));
        JLabel title = new JLabel("ONLINE BANKING", SwingConstants.CENTER);
        title.setFont(new Font("SansSerif", Font.BOLD, 24));
        main.add(title, BorderLayout.NORTH);
        JPanel form = new JPanel(new GridLayout(4, 2, 10, 10));
        form.add(new JLabel("Recipient Account No.:"));
        form.add(accountField);
        form.add(new JLabel("Transfer Amount (Rs):"));
        form.add(amountField);
        JButton transferBtn = new JButton("Transfer Funds");
        JButton balanceBtn = new JButton("Balance Inquiry");
        JButton historyBtn = new JButton("Transaction History");

        JButton clearBtn = new JButton("Clear");

        form.add(transferBtn);
        form.add(balanceBtn);
        form.add(historyBtn);
```

```
form.add(clearBtn);
```

SOURCE CODE (CONTINUED)

```
main.add(form, BorderLayout.CENTER);
balanceLabel.setFont(new Font("SansSerif", Font.BOLD, 16));
main.add(balanceLabel, BorderLayout.SOUTH);
historyArea.setEditable(false);
historyArea.setRows(6);
transferBtn.addActionListener(e -> transferFunds());
balanceBtn.addActionListener(e ->
    JOptionPane.showMessageDialog(this,
        String.format("Available Balance: Rs%.2f", balance),
        "Balance Inquiry",
        JOptionPane.INFORMATION_MESSAGE));
historyBtn.addActionListener(e -> showHistory());
clearBtn.addActionListener(e -> {
    accountField.setText("");
    amountField.setText("");
});
add(main);
setVisible(true);
}

private void transferFunds() {
    String account = accountField.getText().trim();
    String amountText = amountField.getText().trim();
    if (!account.matches("\\d{10,16}")) {
        JOptionPane.showMessageDialog(this,
            "Invalid account number. Enter 10-16 digits.",
            "Validation Error", JOptionPane.ERROR_MESSAGE);
        return;
    }
    double amount;
    try {
        amount = Double.parseDouble(amountText);
    } catch (NumberFormatException ex) {
        JOptionPane.showMessageDialog(this,
            "Invalid amount. Enter a numeric value.",
            "Validation Error", JOptionPane.ERROR_MESSAGE);
        return;
    }
    if (amount <= 0) {
        JOptionPane.showMessageDialog(this,
            "Amount must be greater than zero.",
            "Validation Error", JOptionPane.ERROR_MESSAGE);
        return;
    }
    if (amount > balance) {
        JOptionPane.showMessageDialog(this,
            "Transaction failed: insufficient balance.",
            "Transaction Error", JOptionPane.ERROR_MESSAGE);
        return;
    }
    balance -= amount;
    transactions.add("Transferred Rs" + String.format("%.2f", amount)
        + " to A/C " + maskAccount(account));
    balanceLabel.setText(String.format("Balance: Rs%.2f", balance));
    JOptionPane.showMessageDialog(this,
        String.format("Transfer successful.%nRemaining balance: Rs%.2f", balance),

        "Success", JOptionPane.INFORMATION_MESSAGE);
    accountField.setText("");
    amountField.setText("");
}

private String maskAccount(String account) {
    return "*****" + account.substring(account.length() - 4);
}

private void showHistory() {
    historyArea.setText("");
    if (transactions.isEmpty()) {
        historyArea.append("No transactions available.");
    } else {
        for (String t : transactions) {
            historyArea.append(t + "\n");
        }
    }
}
```

SOURCE CODE (CONTINUED)

```
        }
    }
    JOptionPane.showMessageDialog(this,
        new JScrollPane(historyArea),
        "Transaction History",
        JOptionPane.INFORMATION_MESSAGE);
}
public static void main(String[] args) {
    SwingUtilities.invokeLater(OnlineBankingInterface::new);
}
}
```